Shubham Gupta

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Summary

Software Engineer specializing in backend architecture, full-stack development, and AI integration. Proven track record of engineering scalable RESTful APIs, implementing microservices with Spring Boot, and optimizing complex database architectures. Adept at driving end-to-end system design, enhancing application security with JWT, and leveraging NLP and OCR pipelines to deliver high-performance, production-ready enterprise solutions.

Experience

Software Engineer, Zensar Technologies - Kolkata, WB — Mar 2026 – Present

- Engineered modular and scalable RESTful APIs using Spring Boot, reducing service coupling and improving system maintainability across distributed architectures.
- Architected and implemented JWT-based authentication and authorization flows, securing microservice endpoints and establishing robust identity verification.
- Orchestrated API Gateway routing and seamless integration patterns, optimizing inter-service communication and overall backend performance.
- Led technical execution spanning the full stack (React.js, Java, TypeScript), maintaining high availability and efficient data flow across MySQL and MongoDB databases.

Software Intern, Damodar Valley Corporation (DVC) - Kolkata, WB — July 2024 – Sept 2024

- Developed an OCR-powered search engine for scanned circulars with 92% accuracy, reducing document retrieval latency by 35%.
- Built a citation-based structured indexing system, enhancing document accessibility and efficiency for enterprise compliance workflows.

Data Analyst Intern, IBM SkillsBuild - Remote — June 2023 – July 2023

- Designed *HealthTrack*, a disease prediction model utilizing machine learning and Python (Streamlit), achieving 85% prediction accuracy.
- Applied advanced analytics and data visualization (Tableau, Power BI) to uncover actionable insights, improving clarity of healthcare reporting by 20%.

Skills

Domains & Architecture: Full Stack Development, Java Backend Development, Microservices, REST APIs, Event-Driven Architecture, AI/ML, System Design
Programming Languages: Java, Python, JavaScript, TypeScript, C
Frameworks & Libraries: Spring, Spring Boot, Spring Data JPA, Spring Security, Spring Batch, React.js, Angular, Kafka
Databases: MySQL, MongoDB, PostgreSQL
Tools & Platforms: Git, Docker, Maven, Jenkins, Postman, Azure, STS, IntelliJ IDEA
Testing & Issue Tracking: Unit Testing, Mockito, Code Review, Git (Version Control & Issue Handling)
AI & ML Core: TensorFlow, PyTorch, Keras, Scikit-Learn, NLP, Computer Vision, CNN, RNN, YOLO, Numpy, Pandas

Projects

OLX Classified Platform (Microservices Architecture)

- Engineered a classified advertisements platform enabling users to securely post, manage, and browse advertisements using Java, Spring Boot, and React.js.
- Architected scalable microservices to handle authentication and advertisement workflows, incorporating JWT for secure API communication.
- Implemented service integration, centralized request handling via API Gateway, and optimized database interactions using MySQL and MongoDB.

AI-Powered Chat Assistant for Corporate Documentation — Documentation

- Designed a scalable chatbot using Azure Cognitive Services, AI Search, and Cosmos DB for dynamic document retrieval.
- Integrated OCR and NLP pipelines with RAG architecture, improving overall accessibility and reducing data retrieval time by 35%.

VyomAI – Multimodal AI Chatbot — VyomAI - GitHub Repo

- Built a high-performance chatbot integrating natural language processing and vision models for contextual Q&A automation.

InsurAI – Corporate Policy Automation System

- Developed a full-stack web application for automated insurance policy management using Spring Boot and React.js.
- Built secure RESTful APIs with optimized MySQL schemas and integrated AI-driven modules for intelligent policy recommendation.

Education

MCKV Institute of Engineering (MCKVIE), B.Tech in Artificial Intelligence & Machine Learning — Sept 2021 – June 2025

- CGPA: 9.37/10

Maria's Day School, Council for Indian School Certificate Examinations — July 2021

- 12th Percentage: 82.67 | 10th Percentage: 76.80

Achievements

- Winner – ACM India East & N.E. Dissertation Contest (2024).
- Winner – Intra-Institutional Business Plan Competition, MCKVIE (2025).
- 2nd Place – Innovative Idea Contest, Pragati, MCKVIE (2024).